

Comparing morphological and phonological cues in grammatical gender assignment in French as a second language

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Background

Noun endings may provide cues for grammatical gender assignment in French, but some phonological and morphological cues clash.

- French has a bipartite system (masculine/feminine)
- In addition to other elements, gender is marked on singular determiners (*le/un; la/une*):
(1) *le/un* chapeau (2) *la/une* pomme
the_{MASC-SG} hat the_{FEM-SG} apple

What about the role of noun endings in acquisition?

- L1 children are sensitive to endings (e.g., Karmiloff-Smith, 1979) **VS.** it's a masculine-default pattern (Boloh et al., 2013)
- (Adult) L2 learners over-rely on phonological cues (Hawkins & Franceschina, 2004) **VS.** they are not sensitive to morphophonological cues (Meisel et al., 2018).
- Some adult bilinguals are less sensitive to cues from derivational morphology (compared to final segments), (Kupisch et al., 2013).

Clashing cues: phonology vs. semantics

Bilingual children were more sensitive to phonological than semantic cues when these cues clashed (Kupisch et al., 2022).

Correlations of varying strength exist between gender and noun endings.

Compare (3-4):

An arguably reliable cue

(3) Nouns ending in a vowel tend to be masculine (e.g., Harley, 1979).

The case of final [ɔ]

(4) Final [ɔ] is not reliably masculine (30%) when simple and complex words are grouped together (Tucker et al., 1977) but is reliably masculine (100%) when only monosyllabic words are analyzed (Kupisch et al., 2002).

What exactly constitutes a "noun ending" for gender cues (Carroll, 1999)??

Morphological and phonological cues can be in conflict (cf., 5-6). Derivational cues are deterministic.

(5) final [ɔ] in simple (a) vs. complex words -ation (b)

a. ballon_{MASC} [balɔ̃]
ball
b. créer + -ation_{FEM} [asjɔ̃] -> création_{FEM}
create_{INF} creation

(6) Some compounds are masculine, even if they include feminine nouns

(6) V-N compounds with verbal heads
rincer + bouche_{FEM} -> rince-bouche_{MASC}
rinse_{INF} mouth 'mouth rinse' [ʁɛ̃sbuʃ]

Research questions

- To what extent are adult L2 learners sensitive to final segments versus derivational morphology when assigning gender?
- Does this sensitivity change when these cues are in conflict (e.g., monomorphemic nouns ending in [ɔ] tend to be masculine, but deverbal nouns with -ation [asjɔ̃] are feminine)?

Method

20 adult L2 French learners (L1 English university students) took a metalinguistic grammatical gender identification task.

Task description: Learners saw a noun on a computer screen and were asked to say (1) whether they knew the word (*yes, not sure, no*), (2) whether it was masculine or feminine, and (3) whether they were confident in their answer (4 pt scale). Words that participants claimed not to know were excluded from the analysis.

Predictions

- If learners are sensitive to morphological properties when assigning gender, then they should perform better on complex compared to simple words.
- If learners are more sensitive to the final segment, then there should be patterns in the clash condition that are not present in the no-clash condition.
- If limited sensitivity to either final segments or derivational suffixes, then patterns will be similar across clash and no-clash conditions.

Stimuli

Clash conditions			No-clash conditions		
	MASC	FEM		MASC	FEM
morphologically	final [ɔ]	final [j]	morphologically	final [ɑ̃]	final [z]
simple	<i>n</i> = 6	<i>n</i> = 6	simple	<i>n</i> = 6	<i>n</i> = 6
Example	<i>le ballon</i> 'ball' <i>la bouche</i> 'mouth'		Example	<i>le gant</i> 'the glove'	<i>la chose</i> 'the thing'
	FEM	MASC		MASC	FEM
morphologically	-ation [asjɔ̃]	Compound nouns	morphologically	-ment [mɑ̃]	-ise [iz]
complex	<i>n</i> = 6	ending in [j] and headed by verbs <i>n</i> = 6	complex	<i>n</i> = 6	<i>n</i> = 6
Example	<i>la création</i> 'creation'	<i>le rince-bouche</i> 'mouth-rinse'	Example	<i>le bâtiment</i> 'the building'	<i>la bêtise</i> 'the mistake'

Main findings and interpretation

- Morphologically **complex nouns < simple nouns**, especially with compounds, suggesting **little sensitivity to derivational cues**.
- Clash ≈ no-clash, suggesting learners are **not conflating the final segment of a derivational morpheme with the final segment of a simple noun** when assigning gender. Compounds are an exception (see below).
- Overall, there is **limited evidence for sensitivity to morphological or phonological cues**.

The effect of clashing (clash vs. no clash) was not significant, but there was a significant 3-way interaction between clashing, word type (simple vs. complex), and gender. Performance was significantly weaker with masculine complex nouns in the clash condition, as shown in Figure 1.

Fig 1. Clash conditions (% correct)

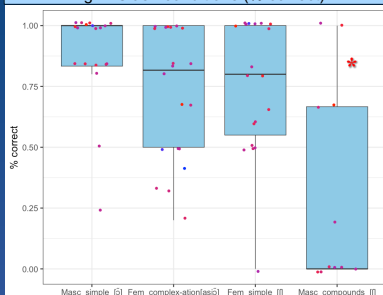
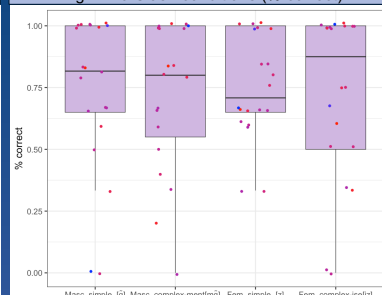


Fig 2. No Clash conditions (% correct)



proficiency



Compounds: evidence for lexical learning?

Most (11/20) learners assigned the same gender to nouns such as *bouche* 'mouth' (fem. simple) as to masculine compounds (*rince-bouche*), which are masculine due to compounding -> learners were sensitive to endings if a whole lexical item was involved.

Future Directions

What kind of training might be efficient for improving retention of gender assignment in classroom learners? -> Spaced retrieval (e.g., Leonard et al., 2019) with canonical nouns to examine effects of retrieval and cue reliability. Repeated spaced retrieval has led to some improvement in word learning in children with DLD, but unclear if this extends to other lexical properties.

Selected References

- Boloh, Y., & Ibberson, L. (2013). Natural gender, phonological cues and the default grammatical gender in French children. *First Language*, 33(5), 449-468.
- Hawkins, R., & Franceschina, F. (2004). Explaining the acquisition and non-acquisition of determiner-noun gender concord in French and Spanish. In P. Prevost & J. Paradis (Eds.), *The Acquisition of French in Different Contexts: Focus on Functional Categories* (pp. 175-206). John Benjamins Publishing Company.
- Karmiloff-Smith, A. (1979). *A functional approach to child language: A study of determiners and reference*. Cambridge University Press.
- Kupisch, T., Akpinar, D., & Stöhr, A. (2013). Gender assignment and gender agreement in adult bilinguals and second language learners of French. *Linguistic Approaches to Bilingualism*, 3(2), 150-179.
- Kupisch, T., Müller, N., & Cantone, K. F. (2002). *Gender in monolingual and bilingual first language acquisition: Comparing Italian and French*.
- Meisel, J. M. (2018). Early child second language acquisition: French gender in German children. *Bilingualism: Language and Cognition*, 21(4), 656-673.

$R^2 = 0.41$, $p < .001$, AIC = 262, BIC = 288, Log-likelihood = -124

$R^2 = 0.20$, $p < .001$, AIC = 364, BIC = 391, Log-likelihood = -175